DISPROOF OF EVENTUAL EQUALITY IN ERDŐS PROBLEM 878 (QUESTION 3B)

ILIA PAUZNER

For $n \in \mathbb{N}$, let $\mathcal{P}(n)$ denote the set of prime factors of n (so $\mathcal{P}(0) = \emptyset$), and for a prime p let $q_p(n) := p^{\lfloor \log_p n \rfloor}$ (with $\lfloor \log_p 0 \rfloor := 0$), which for $1 \le n$ is the largest power of p not exceeding n. Erdős [1] defined

$$f(n) := \sum_{p \in \mathcal{P}(n)} q_p(n), \qquad F(n) := \max \sum_{a \in A} a,$$

where $A \subseteq [2, n]$ ranges over finite sets of pairwise coprime integers a with $\mathcal{P}(a) \subseteq \mathcal{P}(n)$, and asked whether the maxima $\operatorname{maxUpTo} f\, x := \max_{n \le x} f(n)$ and $\operatorname{maxUpTo} F\, x := \max_{n \le x} F(n)$ satisfy $\operatorname{maxUpTo} f\, x = \operatorname{maxUpTo} F\, x$ for all sufficiently large x.

Theorem. *For every $X_0 \in \mathbb{N}$, there exists $x \ge X_0$ such that* $\operatorname{maxUpTo} f\, x < \operatorname{maxUpTo} F\, x$.

Consequently, it is not true that $\operatorname{maxUpTo} f\, x = \operatorname{maxUpTo} F\, x$ for all sufficiently large x, which answers Question 3b negatively.

Correspondence with the Lean formalization. The proof below is the one formalized in `FormalConjectures/ErdosProblems/878.lean`, step by step, with the same names ($q_1, \dots, q_6$, Q_6, C_6, N_{smooth}, N_2, A, M, step, rem, K, Y_0, Y_a, Y_b, Q_a, Q_b, R', D, r_a, r_b, b_i, J, ℓ, z_j, …) and with every inequality in the form used in Lean. All variables are natural numbers, except $\operatorname{rem}(k)$, which is real. The remaining correspondences are:

This note	Lean
$\mathcal{P}(n)$; $\lfloor \log_p n \rfloor$; $q_p(n)$	`n.primeFactors`; `Nat.log p n`; `q_p p n` (in Lemma 2, $q_2(\cdot)$ is `q_p 2`, not the constant `q2`)
$\lfloor a/b \rfloor$; $[a, b]$; $\{0, \dots, M-1\}$	`a / b` in $\mathbb{N}$; `Finset.Icc a b`; `Finset.range M`
$\operatorname{maxUpTo} f\, x$	`maxUpTo f x` (maximum over $n \in \{0, \dots, x\}$)
an integer $a \in [2, n]$ with $\mathcal{P}(a) \subseteq P$	`IsAdmissibleTerm n P a`
$14 q_p(\lfloor 21Y/20 \rfloor) < Y$	`IsGoodForPrime p Y`
$q_1, \dots, q_6$ are six distinct primes; $0 < C_6$	`Q6_prime`, `Q6_card`; `C6_pos`
Theorem; the sentence after it	`erdos_878_q3_disproved`; `erdos_878.parts.iii_eventually`
Lemma 1	`lemma1_exists_smooth_gt`; the choice of j is `exists_step_crossing`
inequality (1)	`exists_F_ge_of_smooth`, via `q_p_admissible` and `sum_le_F_of_disjoint`
Lemma 2	`lemma2_f_two_mul_gt`
Lemma 3	`lemma3_superlinear_maxUpTo_f`
Lemma 4	`lemma4_exists_good_Y`
contradiction hypothesis	`h_no_jump`
Step 1	`single_step_jump`; the decomposition of $f(n)$ is `sum_sdiff_erase_two_le`, and $r_a r_b \prod_{p \in R'} p \le n$ is `mul_mul_prod_erase_le`
Step 2: iterating Step 1; rem; $20(z + \operatorname{rem}(k)) \le 21Y \Rightarrow z \le \lfloor 21Y/20 \rfloor$; the induction on k	`iter`; `rem`; `le_cap`; `h_seq`
monotonicity of q_p and maxUpTo; $g(n) \le \operatorname{maxUpTo} g\, x$ for $n \le x$ $(g = f, F)$	`q_p_mono`, `maxUpTo_mono`; `le_maxUpTo`

The Lean file states Lemma 2 before Lemma 1, and Step 1 as the separate lemma `single_step_jump` before Lemmas 3 and 4; its hypotheses beyond those of Step 1 (inequality (1), the contradiction hypothesis for $x \geq Y$, $20 \leq Y$, $2N_{\text{smooth}} \leq Y$, and the conclusion of Lemma 4 for Y) are the standing assumptions of Step 1 here.

Lemma 1. *If $p < r$ are primes with $98r < 100p$, then for every n with $p^{p^2+2} \leq n$ there exists an integer $z \in [2, n]$ with $\mathcal{P}(z) \subseteq \{p, r\}$ and $98n < 100z$.*

Proof. Put $J = p^2 + 1$ and $\ell = \lfloor \log_p n \rfloor$; then $J < \ell$. Let $z_j = p^{\ell-j} r^j$. Then $z_0 \leq n$, as $z_0 = q_p(n)$, while

$$n < p^{\ell+1} = p^{\ell-J}(p^J p) \leq p^{\ell-J}(p+1)^J \leq z_J,$$

since $p^J p \leq p^J + J p^{J-1} \leq (p+1)^J$ (the first inequality because $J = p^2 + 1$). Choose $j < J$ with $z_j \leq n$ and $n < z_{j+1}$. Since $z_j r = z_{j+1} p$, we have $98np < 98 z_{j+1} p = 98 z_j r < 100 z_j p$, so $98n < 100 z_j$; as $2 \leq n$, also $2 \leq z_j$. Hence $z = z_j$ has the required properties. □

Fix the six primes

$$q_1 := 168000061, \quad q_2 := 168000101, \quad q_3 := 168000121,$$
$$q_4 := 168000139, \quad q_5 := 168000143, \quad q_6 := 168000167,$$

let $Q_6 := \{q_1, \ldots, q_6\}$, let $C_6 := \prod_{p \in Q_6} p$, and let $N_{\text{smooth}} := q_1^{q_1^2+2} + q_3^{q_3^2+2} + q_5^{q_5^2+2}$. For every m with $N_{\text{smooth}} \leq m$, Lemma 1 applies to the pairs (q_1, q_2), (q_3, q_4), and (q_5, q_6) (as $q_{2i-1} < q_{2i}$, $98 q_{2i} < 100 q_{2i-1}$, and $q_{2i-1}^{q_{2i-1}^2+2} \leq N_{\text{smooth}} \leq m$) and gives $b_1, b_2, b_3 \in [2, m]$ with $\mathcal{P}(b_i) \subseteq \{q_{2i-1}, q_{2i}\}$ and $98m < 100 b_i$. If moreover $R' \subseteq \mathcal{P}(m)$, $R' \cap Q_6 = \emptyset$, and $Q_6 \subseteq \mathcal{P}(m)$, then b_1, b_2, b_3 and $q_p(m) \in [2, m]$ ($p \in R'$) are pairwise coprime with prime factors dividing m, so $b_1 + b_2 + b_3 + \sum_{p \in R'} q_p(m) \leq F(m)$ and hence

$$294m + 100 \sum_{p \in R'} q_p(m) < 100 F(m). \tag{1}$$

Lemma 2. *For all n with $1 \leq n$, $f(n) < f(2n)$.*

Proof. We have $\mathcal{P}(n) \subseteq \mathcal{P}(2n)$, $2 \in \mathcal{P}(2n)$, and $q_p(n) \leq q_p(2n)$ for $p \in \mathcal{P}(n)$. If $2 \in \mathcal{P}(n)$, then $q_2(n) < q_2(2n)$, as $\lfloor \log_2(2n) \rfloor = \lfloor \log_2 n \rfloor + 1$; hence $f(n) < \sum_{p \in \mathcal{P}(n)} q_p(2n) \leq f(2n)$. If $2 \notin \mathcal{P}(n)$, then $f(n) \leq \sum_{p \in \mathcal{P}(n)} q_p(2n) < f(2n)$, since $f(2n)$ also contains the term $q_2(2n)$ and $0 < q_2(2n)$. □

Lemma 3. *For every $R \in \mathbb{N}$, there is N_2 such that $Ry < \operatorname{maxUpTo} f\, y$ for all $y \geq N_2$.*

Proof. Since $\sum_p p^{-1} = \infty$, choose N with $2R + 1 \leq \sum_{p<N} p^{-1}$, and set $P = \prod_{p<N} p$ and $N_2 = 2P$. For $y \geq N_2$, let $m = P\lfloor y/P \rfloor$; then $m \leq y$ and $y < 2m$. Every prime $p < N$ divides m and satisfies $mp^{-1} \leq q_p(m)$, so

$$(2R+1)m \leq m \sum_{p<N} p^{-1} \leq \sum_{p<N} q_p(m) \leq \sum_{p \in \mathcal{P}(m)} q_p(m) = f(m) \leq \operatorname{maxUpTo} f\, y,$$

and $y < 2m$ gives $Ry < \operatorname{maxUpTo} f\, y$. □

Lemma 4. *For every X_0, there is Y with $\max(X_0, 20) \leq Y$ and $14 q_p(\lfloor 21Y/20 \rfloor) < Y$ for all $p \in Q_6$.*

Proof. Let $Y_0 = \max(X_0, 20)$, and for $p \in Q_6$ let $\operatorname{bad}(p)$ be the set of $m < 7$ for which $14 q_p(\lfloor 21 Y_0 15^m/20 \rfloor) < Y_0 15^m$ fails. We show $|\operatorname{bad}(p)| \leq 1$. Let $m_1 < m_2$ lie in $\operatorname{bad}(p)$, and put $Y_a = Y_0 15^{m_1}$, $Y_b = Y_0 15^{m_2}$, $Q_a = q_p(\lfloor 21 Y_a/20 \rfloor)$, and $Q_b = q_p(\lfloor 21 Y_b/20 \rfloor)$. Then $168000000 \leq p$, $20 \leq Y_a$, $15 Y_a \leq Y_b$, $Y_b \leq 11390625\, Y_a$, $Q_a \leq \lfloor 21 Y_a/20 \rfloor$ and $Q_b \leq \lfloor 21 Y_b/20 \rfloor$ (so $20 Q_a \leq 21 Y_a$

and $20Q_b \le 21Y_b$), $Y_a \le 14Q_a$, and $Y_b \le 14Q_b$. Hence $280Q_a \le 294Y_a < 300Y_a \le 20Y_b \le 280Q_b$, so $Q_a < Q_b$, and since Q_a and Q_b are powers of p, $Q_a p \le Q_b$. Therefore

$$\begin{aligned} 3360000000\,Y_a = 20 \cdot 168000000\,Y_a &\le 280\,p\,Q_a \le 280\,Q_b \\ &\le 294\,Y_b \le 294 \cdot 11390625\,Y_a = 3348843750\,Y_a, \end{aligned}$$

which contradicts $20 \le Y_a$. Thus $\sum_{p\in Q_6} |\mathrm{bad}(p)| \le 6 < 7$, so some $m < 7$ lies in no $\mathrm{bad}(p)$, and $Y = Y_0 15^m$ satisfies $\max(X_0, 20) \le Y$ and $14q_p(\lfloor 21Y/20\rfloor) < Y$ for all $p \in Q_6$. □

Proof of the Theorem. Suppose for contradiction that $\mathrm{maxUpTo}\,F\,x \le \mathrm{maxUpTo}\,f\,x$ for all $x \ge X_0$. Put $A := 2000C_6 + 1$. By Lemma 3 with $R = 2(A+10)$, there is N_2 such that $2(A+10)y < \mathrm{maxUpTo}\,f\,y$ for all $y \ge N_2$. Let Y be given by Lemma 4 applied to $\max(X_0, N_2, 2N_{\mathrm{smooth}})$; then $X_0 \le Y$, $N_2 \le Y$, $20 \le Y$, $2N_{\mathrm{smooth}} \le Y$, and $14q_p(\lfloor 21Y/20\rfloor) < Y$ for all $p \in Q_6$.

Step 1. Suppose $Y \le y$, $y \le \lfloor 21Y/20\rfloor$, $1 \le M$, and $105(M+8)Y \le 100\,\mathrm{maxUpTo}\,f\,y$. Choose $n \le y$ with $\mathrm{maxUpTo}\,f\,y = f(n)$.

- $0 < n$: otherwise $\mathrm{maxUpTo}\,f\,y = f(0) = 0$, contradicting $105(M+8)Y \le 100\,\mathrm{maxUpTo}\,f\,y$ and $20 \le Y$. If $2n \le y$, then $f(2n) \le \mathrm{maxUpTo}\,f\,y = f(n)$, contradicting Lemma 2; hence $y < 2n$.
- Since $f(n) \le n\,|\mathcal{P}(n)|$ and $20n \le 21Y$, we have $105(M+8)Y \le 100f(n) \le 100n\,|\mathcal{P}(n)| \le 105Y\,|\mathcal{P}(n)|$, so $M + 8 \le |\mathcal{P}(n)|$.
- Since $|\mathcal{P}(n)| - |Q_6| \le |\mathcal{P}(n) \setminus Q_6|$, $|\mathcal{P}(n) \setminus Q_6| - M \le |(\mathcal{P}(n) \setminus Q_6) \setminus \{0, \dots, M-1\}|$, $|Q_6| = 6$, and $M + 8 \le |\mathcal{P}(n)|$, we get $1 < |(\mathcal{P}(n) \setminus Q_6) \setminus \{0, \dots, M-1\}|$. Take distinct r_a, r_b in this set, so that $r_a, r_b \in \mathcal{P}(n) \setminus Q_6$, $M \le r_a$, and $M \le r_b$.
- Put $R' := (\mathcal{P}(n) \setminus Q_6) \setminus \{r_a, r_b\}$ and $D := \big(\prod_{p\in R'} p\big)C_6$. Since $r_a r_b \prod_{p\in R'} p$ divides n, we have $r_a r_b \prod_{p\in R'} p \le n$; together with $M^2 \prod_{p\in R'} p \le r_a r_b \prod_{p\in R'} p$ and $n \le 2Y$ (as $n \le y \le \lfloor 21Y/20\rfloor$), this gives $DM^2 \le 2C_6 Y$, that is,

$$D \le \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor.$$

- Let $m := D(\lfloor n/D\rfloor + 1)$; then $n < m$ and $m \le y + D$. Since $D \mid m$, we have $R' \subseteq \mathcal{P}(m)$ and $Q_6 \subseteq \mathcal{P}(m)$; also $R' \cap Q_6 = \emptyset$ and $N_{\mathrm{smooth}} \le m$, as $2N_{\mathrm{smooth}} \le Y \le y < 2n < 2m$. So (1) gives $294m + 100\sum_{p\in R'} q_p(m) < 100F(m)$. Next, $\sum_{p\in R'} q_p(n) \le \sum_{p\in R'} q_p(m)$, and since every prime factor of n lies in $\{r_a, r_b\} \cup R' \cup Q_6$,

$$f(n) \le q_{r_a}(n) + q_{r_b}(n) + \sum_{p\in R'} q_p(n) + \sum_{q\in Q_6} q_q(n),$$

where $q_{r_a}(n) \le n$ and $q_{r_b}(n) \le n$. Each $q \in Q_6$ satisfies $14q_q(n) \le 14q_q(\lfloor 21Y/20\rfloor) < Y \le y < 2n$, that is, $7q_q(n) < n$; summing over the six elements of Q_6 gives $7\sum_{q\in Q_6} q_q(n) < 6n$. Hence $7f(n) < 20n + 7\sum_{p\in R'} q_p(m)$, and, as $n < m$,

$$700f(n) + 58n < 2058m + 700\sum_{p\in R'} q_p(m) < 700F(m).$$

As $29Y \le 29y < 58n$, this gives $700f(n) + 28Y < 700F(m)$, that is, $25f(n) + Y < 25F(m)$. Finally, $F(m) \le \mathrm{maxUpTo}\,F\,(y+D) \le \mathrm{maxUpTo}\,f\,(y+D)$, by $m \le y + D$ and the contradiction hypothesis (as $X_0 \le Y \le y + D$). Hence $y' := y + D$ satisfies

$$y \le y', \qquad y' \le y + \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor, \qquad 25\,\mathrm{maxUpTo}\,f\,y + Y \le 25\,\mathrm{maxUpTo}\,f\,y'. \tag{2}$$

Step 2. Iterating Step 1 gives: if $Y \le y$, $1 \le M$, $105(M+8)Y \le 100\,\mathrm{maxUpTo}\,f\,y$, and $y + j\lfloor 2C_6 Y/M^2\rfloor \le \lfloor 21Y/20\rfloor$, then there is $y' \ge y$ with $y' \le y + j\lfloor 2C_6 Y/M^2\rfloor$ and $25\,\mathrm{maxUpTo}\,f\,y + jY \le 25\,\mathrm{maxUpTo}\,f\,y'$. This follows by induction on j: for $j = 0$ take $y' = y$; for $j + 1$, the y' obtained for j satisfies $Y \le y'$, $y' \le y + j\lfloor 2C_6 Y/M^2\rfloor \le \lfloor 21Y/20\rfloor$, and $105(M+8)Y \le 100\,\mathrm{maxUpTo}\,f\,y \le 100\,\mathrm{maxUpTo}\,f\,y'$, so Step 1 applies to it.

Let $\mathrm{rem}(k) := \frac{54C_6Y}{A+k}$, and note that $20(z+\mathrm{rem}(k)) \le 21Y$ implies $z \le \lfloor 21Y/20 \rfloor$, since $0 \le \mathrm{rem}(k)$ and $z \in \mathbb{N}$. We construct $y_k \ge Y$ by induction on k with

$$20(y_k + \mathrm{rem}(k)) \le 21Y \quad \text{and} \quad 105(A+1+k+8)Y \le 100\,\mathrm{maxUpTo}\, f\, y_k.$$

For $k = 0$, take $y_0 = Y$: we have $20\,\mathrm{rem}(0) \le Y$ because $20 \cdot 54C_6Y \le (2000C_6+1)Y$, so $20(Y+\mathrm{rem}(0)) \le 21Y$; and $105(A+1+0+8)Y \le 200(A+10)Y < 100\,\mathrm{maxUpTo}\, f\, Y$ because $2(A+10)Y < \mathrm{maxUpTo}\, f\, Y$. Given y_k, set $M = A+1+k$, so that $1 \le M$, and $\mathrm{step} = \lfloor 2C_6Y/M^2 \rfloor$. Since $\mathrm{step} \cdot M^2 \le 2C_6Y$ and $(A+k)(A+k+1) \le M^2$, we have $27 \cdot \mathrm{step} \cdot (A+k)(A+k+1) \le 54C_6Y$, hence $27 \cdot \mathrm{step} \le \mathrm{rem}(k) - \mathrm{rem}(k+1)$ and $20(y_k + 27 \cdot \mathrm{step} + \mathrm{rem}(k+1)) \le 20(y_k + \mathrm{rem}(k)) \le 21Y$, so $y_k + 27 \cdot \mathrm{step} \le \lfloor 21Y/20 \rfloor$. Iterating Step 1 with $j = 27$ (possible as $Y \le y_k$, $1 \le M$, and $105(M+8)Y = 105(A+1+k+8)Y \le 100\,\mathrm{maxUpTo}\, f\, y_k$) gives $y' \ge y_k$ with $y' \le y_k + 27 \cdot \mathrm{step}$ and $25\,\mathrm{maxUpTo}\, f\, y_k + 27Y \le 25\,\mathrm{maxUpTo}\, f\, y'$. Put $y_{k+1} := y'$. Then $Y \le y_{k+1}$, $20(y_{k+1} + \mathrm{rem}(k+1)) \le 21Y$, and

$$\begin{aligned} 105(A+1+(k+1)+8)Y &\le 105(A+1+k+8)Y + 108Y \\ &\le 100\,\mathrm{maxUpTo}\, f\, y_k + 108Y \le 100\,\mathrm{maxUpTo}\, f\, y_{k+1}. \end{aligned}$$

Finally, let $K := \mathrm{maxUpTo}\, f\, \lfloor 21Y/20 \rfloor + 1$. Since $20(y_K + \mathrm{rem}(K)) \le 21Y$, we have $y_K \le \lfloor 21Y/20 \rfloor$, so $\mathrm{maxUpTo}\, f\, y_K \le \mathrm{maxUpTo}\, f\, \lfloor 21Y/20 \rfloor < K$. But $100K \le 105(A+1+K+8)Y \le 100\,\mathrm{maxUpTo}\, f\, y_K$, a contradiction. □

References

[1] P. Erdős, *On two unconventional number theoretic functions and on some related problems*, Topics in classical number theory, Vol. I, II (Budapest, 1981), Colloq. Math. Soc. János Bolyai **34** (1984), 445–458.